OCA OCPP 2.0.1 / 2.1 PLUGFEST & CONFERENCE AUSTRALIA

21–23 October 2026 · Canberra

From grid envelope to bidirectional charging profile

open, standard, end to end.

Home energy management that turns CSIP-AUS dynamic operating envelopes into OCPP 2.1 bidirectional charging profiles that the charging station validates — responding quickly to price and network events, including 1-minute-notice Minimum System Load and Lack of Reserve events.

1 The chain: network operator to EV socket

every component open source

Network operator CSIP-AUS server
ANU BSGIP Cactus v1.2 test harness

→

Open Dynamic Export (ODE)
CSIP-AUS client dynamic envelopes

→

Home Assistant EEMS
Home Assistant Green bridge + CitrineOS-HA

→

CitrineOS CSMS
OCPP 2.1 / 2.0.1 / 1.6 open-source CSMS

→

EVerest → Plugfest stations
charging stations validate profiles

→

Electric vehicle V2G
ISO 15118-20 EV charge + · discharge −

SEP2 · IEEE 2030.5 CSIP-AUS	MQTT limits	REST CitrineOS API	**OCPP 2.1 SetChargingProfile**	ISO 15118-20 DC_BPT

OCPP 2.1 SetChargingProfile · TxProfile · operationMode CentralSetpoint · setpoint / limit / dischargeLimit in W

Standard OCPP 2.1 smart charging — not reliant on the OCPP 2.1 DER control block. Secondary path: SunSpec Modbus 1.2 to a hybrid inverter.

2 Proven on 17 October 2026

ANU BSGIP Cactus CSIP-AUS v1.2 test harness · EVerest DC V2G simulator

16/16	16/16	±20 kW	±25 kW	600 s	100%
LOA-11 import limits	**LOA-11 import limits**	**tracked exactly**	**DC V2G rating**	**schedule-ahead**	**open source**
AC charger · run 74858	DC V2G charger · run 74865	OCPP 2.1 CentralSetpoint	ISO 15118-20 DC_BPT	≤10 OCPP profile periods	every component

Schedule-ahead: the bridge sends the next 600 s of CSIP-AUS controls as OCPP 2.1 profile periods, so each limit takes effect at its start time.
In run 74865 every decrease applied 0.5–0.6 s early from the held profile, including 0 W.

3 One TxProfile, five setpoints

EVerest DC V2G simulator · ISO 15118-20 DC_BPT · ±25 kW rated · 60 s per step

4 What the demo shows

live, on the Plugfest floor

Export-limit clip
during V2G discharge

Import-limit clip
during charging

Load-limit clip
site load envelope

1-minute MSL / LOR
short-notice events

Local congestion
network constraint

Price changes
AEMO spot price

5 Draft AS 5438: three protocols an EEMS must support

SEP2
via CSIP-AUS
IEEE 2030.5 · AS 5385:2023
Primary path

OCPP 2.1
bidirectional EV charging
SetChargingProfile · TxProfile
Primary path

SunSpec Modbus 1.2
inverters
hybrid inverter
Secondary path

AS 5438 function-map timing targets

≤5 s	≤30 s	≤15 s
read / write	active-power limit	comms-loss fail-safe

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Also at the Plugfest as a CSMS under test: CitrineOS on Home Assistant Green, Security Profile 1

github.com/purcell-lab/ocpp-plugfest

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